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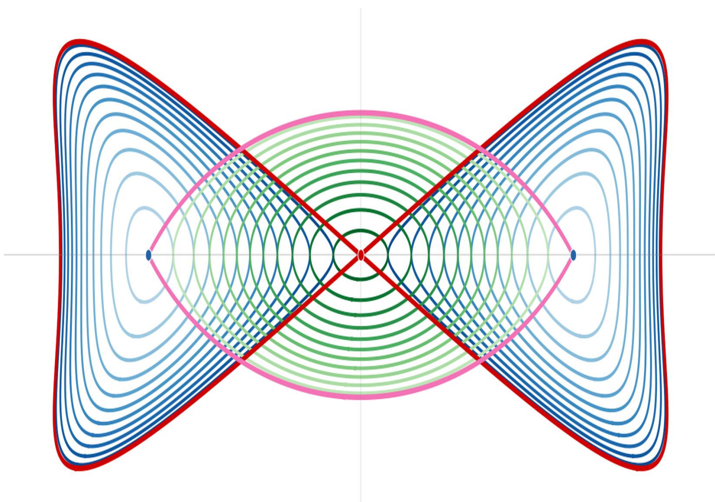
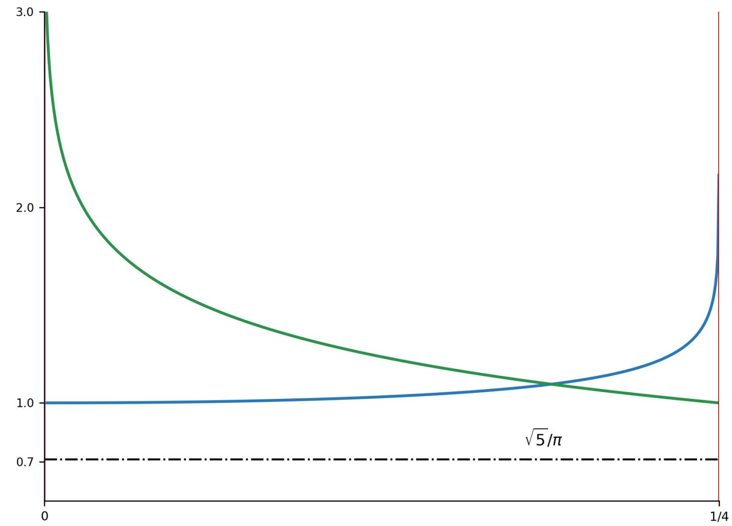


(a) Real and Complex cycles, by Abel–Wick transform.



(b) Periods and Abel's identity.

The μ-stratum $\quad 2H_\mu = p^2 + q^2 + (q^3 - 3p^2 q) + \mu(q^2 - 3p^2)^2, \qquad \alpha = 2H_\mu.$

Transform

$$2H_\mu(p, -2-q) - 2H_\mu(p,q) = -2(4\mu - 1)(q+1)(3p^2 - q^2 - 2q - 2),$$
$$q \mapsto -2 - q \quad \Longrightarrow \quad \mu = \frac{1}{4}, \qquad (p,q) = (0,-1), \qquad \alpha = \frac{1}{4}.$$

Curve

$$\alpha = 2H = p^2 + q^2 + q^3 - 3p^2 q + \frac{(q^2 - 3p^2)^2}{4}, \qquad \rho = (2H)_p = p\big(9p^2 - 3(q+1)^2 + 5\big).$$

Period form

$$\omega = \frac{2\,dq}{\rho} = \frac{dq}{H_p}.$$

Real period

$$T_1(\alpha) = \frac{1}{2\pi}\oint_{\gamma_\alpha^{(1)}} \omega, \qquad T_1(0) = 1, \qquad \gamma_\alpha^{(1)} \subset (p,q) \text{ around } (0,0).$$

Complex period

$$T_2(\alpha) = \frac{\sqrt{5}}{4\pi}\oint_{\gamma_\alpha^{(2)}} i\omega, \qquad T_2(1/4) = 1, \qquad \gamma_\alpha^{(2)} \subset (ip,q) \text{ around } (0,-1).$$

Hypergeometric

$$T_1(\alpha) = (1 + 9\alpha^2)^{-1/4}\,{}_2F_1\left(\frac{1}{12}, \frac{5}{12}; 1; \chi(\alpha)\right), \qquad \chi(\alpha) = -\frac{27\alpha^2(4\alpha - 1)(9\alpha + 4)^2}{4(1 + 9\alpha^2)^3}.$$

Abel's identity

$$\frac{\alpha(3\alpha - 2)(4\alpha - 1)(9\alpha + 4)}{(3\alpha - 2)^2}\left(T_2 T_1' - T_1 T_2'\right) = \frac{\sqrt{5}}{\pi}.$$

Annihilator

$$A_\alpha = P_0 + P_1 \partial_\alpha + P_2 \partial_\alpha^2, \qquad P_0 = 3\alpha(9\alpha - 16),$$
$$P_1 = 216\alpha^3 - 195\alpha^2 - 28\alpha + 8, \qquad P_2 = \alpha(3\alpha - 2)(4\alpha - 1)(9\alpha + 4).$$

Factorization

$$P_1 = P_2' - \frac{6}{3\alpha - 2}P_2, \qquad A_\alpha = (3\alpha - 2)^2 \partial_\alpha\left(\frac{P_2}{(3\alpha - 2)^2}\partial_\alpha\right) + P_0.$$

Certificate

Set $x = q + 1$. Then $\Xi = \dfrac{V(\alpha, p, x)}{\rho^3}$, $\qquad A_\alpha \circ \omega = d\Xi$, where

$$\begin{aligned} V = x\Big[&\alpha^3(486p^2 - 162x^2 + 1080) + \alpha^2(1134p^2x^2 - 405p^2 - 162x^4 + 189x^2 + 105) \\ &+\alpha(-108p^2x^4 - 648p^2x^2 + 1002p^2 + 36x^6 - 156x^4 + 114x^2 + 70) \\ &-108p^2x^4 + 468p^2x^2 - 432p^2 + 36x^6 - 144x^4 + 188x^2 - 80\Big]. \end{aligned}$$

Series

$$T_1(4x) = 1 + 24x^2 + 120x^3 + 2520x^4 + 25200x^5 + 397320x^6 + 5045040x^7 + \cdots,$$
$$T_2(1/4 - 100x) = 1 + 132x + 29220x^2 + 7816080x^3 + 2324945700x^4 + 738712234512x^5 + \cdots.$$

Birational maps $u = q - \sqrt{3}\,p, \quad v = q + \sqrt{3}\,p, \quad A(u) = \dfrac{3u^2+6u+4}{12}, \quad B(u) = \dfrac{u(3u+2)}{6},$

$$x = u+1, \qquad y = 2A(u)v + B(u),$$

$$q = \frac{1}{2}\left[x - 1 + \frac{y - B(x-1)}{2A(x-1)}\right], \qquad p = \frac{1}{2\sqrt{3}}\left[\frac{y-B(x-1)}{2A(x-1)} - (x-1)\right].$$

Jacobi quartic $12y^2 = -x^4 + (2+12\alpha)x^2 + 4\alpha - 1, \qquad \omega = \dfrac{dx}{\sqrt{3}\,y}.$

Quartic period $T_1(\alpha) = \dfrac{{}_2F_1\left(\frac{1}{4},\frac{3}{4};1;z_4\right)}{\sqrt{1+6\alpha}}, \qquad \text{where } z_4 = \dfrac{4\alpha(4+9\alpha)}{(1+6\alpha)^2}.$

Birational again $z_4 = \dfrac{4m}{(1+m)^2}, \qquad a^2, b^2 = 1 + 6\alpha \pm 2\sqrt{\alpha(4+9\alpha)},$

$$k = \frac{a-b}{a+b}, \qquad m = k^2, \qquad U = k\frac{x+b}{x-b}, \qquad V = \frac{\sqrt{3}\,(U-k)^2 y}{b(a-b)},$$

$$x = b\frac{U+k}{U-k}, \qquad y = \frac{b(a-b)}{\sqrt{3}\,(U-k)^2}V.$$

Legendre cubic $V^2 = U(U-1)(U-m), \qquad \omega = -\dfrac{2}{a+b}\dfrac{dU}{V}.$

Cubic period $T_1(\alpha) = \dfrac{2}{a+b}\,{}_2F_1\left(\dfrac{1}{2},\dfrac{1}{2};1;m\right) = \dfrac{4K(k)}{\pi(a+b)}.$

Three periods

$$T_1(\alpha) = \frac{{}_2F_1\left(\frac{1}{4},\frac{3}{4};1;z_4\right)}{\sqrt{1+6\alpha}} = \frac{2}{a+b}\,{}_2F_1\left(\frac{1}{2},\frac{1}{2};1;m\right)$$

$$= \frac{4K(k)}{\pi(a+b)} = \frac{{}_2F_1\left(\frac{1}{12},\frac{5}{12};1;X\right)}{(1+9\alpha^2)^{1/4}}.$$

where $X(\alpha) = -\dfrac{27\alpha^2(4\alpha-1)(9\alpha+4)^2}{4(1+9\alpha^2)^3}.$

Eisenstein root $E_4(\tau)^{1/4} = {}_2F_1\left(\dfrac{1}{12},\dfrac{5}{12};1;X\right), \qquad X = \dfrac{1728}{j(\tau)}, \qquad E_4(i\infty)^{1/4} = 1.$

Parameter chain $m = k^2 = \dfrac{1+6\alpha-\sqrt{1-4\alpha}}{1+6\alpha+\sqrt{1-4\alpha}}, \qquad X = \dfrac{27m^2(1-m)^2}{4(1-m+m^2)^3} = X(\alpha).$

Same fourth root $E_4(\tau)^{1/4} = (1+9\alpha^2)^{1/4}T_1(\alpha) = \dfrac{4K(k)}{\pi(a+b)}(1+9\alpha^2)^{1/4}.$

Algebraic factor $(a+b)^4(1-m+m^2) = 16(1+9\alpha^2), \qquad 1-m+m^2 = 1-k^2k'^2.$

Ramanujan $E_4(\tau) = \left(\dfrac{2K(k)}{\pi}\right)^4\left(1-k^2k'^2\right).$

"9. We may get a still closer approximation from the following results. It is known that

$$1 + 240\sum_{r=1}^{\infty}\frac{r^3q^{2r}}{1-q^{2r}} = \left(\frac{2K}{\pi}\right)^4\left(1-k^2k'^2\right), \qquad \ldots"$$

— S. Ramanujan, *Modular equations and approximations to* π, §9 (1914).